



Yakeen Plus (2025)

Short Practice Test - 05

DURATION: 60 Minutes

DATE: 13/10/2024

M. MARKS: 192

Topics Covered

Physics:	Thermal Properties of Matter; Kinetic Theory; Thermodynamics
Chemistry:	Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Biology:	(Botany) : Plant Kingdom, Morphology of Flowering Plants (Zoology) : Locomotion & Movement

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

1. "A good absorber is a good emitter" is explained by:

(1) Stefan's law
(2) Wien's law
(3) Newton's law of cooling
(4) Kirchhoff's law

2. 80 gm of water at 30°C is poured on a large block of ice at 0°C . The mass of ice that melts is:

(1) 30 gm
(2) 80 gm
(3) 1600 gm
(4) 150 gm

3. For an isobaric process, the ratio of ΔQ (amount of heat supplied) to the ΔW (work done by the gas) is:

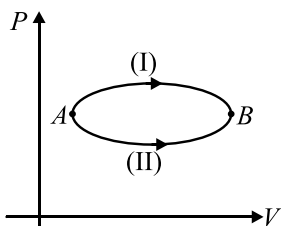
$$\left(\gamma = \frac{C_P}{C_V} \right)$$

(1) γ
(2) $\gamma - 1$
(3) $\frac{\gamma}{\gamma + 1}$
(4) $\frac{\gamma}{\gamma - 1}$

4. On heating a uniform metallic cylinder length increases by 3%. The area of cross-section of its base will increase by:

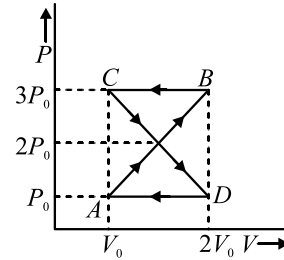
(1) 1.5%
(2) 3%
(3) 9%
(4) 6%

5. A system goes from A to B via two process (I) and (II) as shown in figure. If ΔU_1 and ΔU_2 are change in internal energy in process (I) and (II) respectively then:



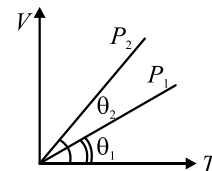
(1) $\Delta U_1 > \Delta U_2$
(2) $\Delta U_1 < \Delta U_2$
(3) $\Delta U_1 = \Delta U_2$
(4) $\Delta U_1 = \Delta U_2 = 0$

6. A thermodynamic system undergoes cyclic process ABCDA as shown in figure. The work done by the system in the cycle is:



(1) $P_0 V_0$
(2) $2P_0 V_0$
(3) $\frac{P_0 V_0}{2}$
(4) Zero

7. In the given $(V - T)$ diagram, what is the relation between pressures P_1 and P_2 ?



(1) $P_2 > P_1$
(2) $P_2 < P_1$
(3) Cannot be predicated
(4) $P_2 = P_1$

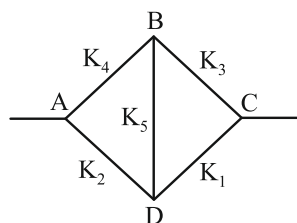
8. 2 Moles of a gas having $\gamma = \frac{7}{5}$ mixed with 1 mole of a gas having $\gamma = \frac{5}{3}$. The γ for the mixture will be:

(1) $\frac{5}{11}$
(2) $\frac{19}{13}$
(3) $\frac{13}{19}$
(4) $\frac{15}{11}$

9. Five molecules of a gas have speeds 1, 2, 3, 4 and 5 km/s. The ratio of rms speed to the average speed of the gas molecule will be:

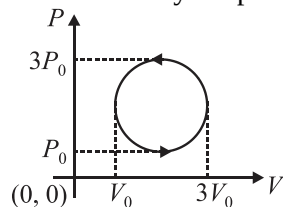
(1) 10 : 3
(2) $\sqrt{11} : 3$
(3) 3 : 10
(4) $\sqrt{3} : \sqrt{10}$

10. Five rods of same dimensions are arranged as shown in the figure. They have thermal conductivities k_1, k_2, k_3, k_4 and k_5 . When points A and C are maintained at different temperature, no heat flows through the central rod if:



- (1) $K_1 = K_4$ and $K_2 = K_3$
- (2) $K_1 K_4 = K_2 K_3$
- (3) $K_1 K_2 = K_3 K_4$
- (4) $\frac{K_1}{K_4} = \frac{K_2}{K_3}$

11. Work done in the cyclic process shown in figure is:



- (1) $4P_0V_0$
- (2) $-4P_0V_0$
- (3) $-\frac{22}{7}P_0V_0$
- (4) $-13P_0V_0$

12. Select the **incorrect** statement.

- (1) A body radiates at all temperatures except 0 K.
- (2) A good reflector is a bad radiator.
- (3) A colder body can radiate heat to the hotter surroundings.
- (4) A body does not radiate when its temperature is below 0°C .

[Short Practice Test-05 | Yakeen Plus (2025) | 13/10/2024]

Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure

SECTION-(II) CHEMISTRY

13. The hybridization state of central iodine atom in IF_5 , I_3^- and I_3^+ are respectively:

- (1) sp^3d^2 , sp^3d and sp^3
- (2) sp^3d , sp^3d and sp^3
- (3) sp^3d^2 , sp^3d^2 and sp^3
- (4) sp^3d , sp^3d and sd^3

14. Among the following elements the configuration having the highest ionization energy is:

- (1) $[\text{Ne}]3\text{s}^23\text{p}^1$
- (2) $[\text{Ne}]3\text{s}^23\text{p}^3$
- (3) $[\text{Ne}]3\text{s}^23\text{p}^2$
- (4) $[\text{Ar}]3\text{d}^{10}4\text{s}^24\text{p}^3$

15. The electron adds to which one of the following orbitals during change of N_2 to N_2^- :

- (1) σ orbital
- (2) σ^* orbital
- (3) π^* orbital
- (4) π orbital

16. What is the **correct** decreasing order of ionic radii of following ions:

- $\text{N}^{3-}, \text{O}^{2-}, \text{F}^-, \text{Na}^+, \text{Mg}^{2+}$
- (1) $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Mg}^{2+} > \text{Na}^+$
 - (2) $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+}$
 - (3) $\text{N}^{3-} > \text{O}^{2-} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^-$
 - (4) $\text{Na}^+ > \text{F}^- > \text{O}^{2-} > \text{Mg}^{2+} > \text{N}^{3-}$

17. The most covalent halide is:

- (1) AlF_3
- (2) AlCl_3
- (3) AlBr_3
- (4) AlI_3

18. Choose the **incorrect** order with respect to the given physical property.

- (1) $\text{Na} > \text{Na}^+$ (Size)
- (2) $\text{N} > \text{C}$ (Electronegativity)
- (3) $\text{Ga} > \text{Al}$ (Metallic nature)
- (4) $\text{Ne} > \text{Ar}$ (Ionization Enthalpy)

19. Match List-I with List-II.

List-I (Compounds)		List-II (Shape of Compounds)	
(A)	SF_6	(I)	Octahedral
(B)	NH_3	(II)	T-shape
(C)	BF_3	(III)	Pyramidal
(D)	ClF_3	(IV)	Trigonal planar

Choose the **correct** answer from the options given below.

- (1) A-I, B-III, C-IV, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-IV, B-I, C-II, D-III

20. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Shielding effect decrease as we go down the group.

Reason R: More is the electrons in the penultimate shell, more is shielding.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

21. Three diatomic species are listed below in different sequences. Which of these presents the **correct** order of their increasing bond order:

- (1) $C_2^{2-} < He_2^+ < O_2^-$
- (2) $He_2^+ < O_2^- < C_2^{2-}$
- (3) $O_2^- < C_2^{2-} < He_2^+$
- (4) $C_2^{2-} < O_2^- < He_2^+$

22. Atomic number of element Ununnilium is:

- (1) 101 (2) 110
- (3) 111 (4) 100

23. The electronic configuration of the element with maximum electron affinity is:

- (1) $1s^2, 2s^2, 2p^3$
- (2) $1s^2, 2s^2, 2p^5$
- (3) $1s^2, 2s^2, 2p^6, 3s^2, 3p^5$
- (4) $1s^2, 2s^2, 2p^6, 3s^2, 3p^3$

24. Which of the following is the **correct** order of dipole moment?

- (1) $NH_3 < BF_3 < NF_3 < H_2O$
- (2) $BF_3 < NF_3 < NH_3 < H_2O$
- (3) $BF_3 < NH_3 < NF_3 < H_2O$
- (4) $H_2O < NF_3 < NH_3 < BF_3$

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Plant Kingdom, Morphology of Flowering Plants

SECTION-(III) BOTANY

25. China rose and mustard both have:

- (1) zygomorphic flowers.
- (2) flowers with superior ovary.
- (3) bisexual epigynous flowers.
- (4) opposite phyllotaxy.

26. Which of the following statement is **correct** w.r.t. vexillary aestivation?

- (1) Largest petals are wings
- (2) Smallest petals are found posteriorly
- (3) Wings are smallest petals
- (4) Anterior petals are keel

27. Given below are two statements:

Statement I: Leaves of dicotyledonous plants generally possess reticulate venation.

Statement II: A bud is present in the axil of leaflets of the compound leaf.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but statement II is incorrect.
- (2) Statement I is incorrect but statement II is correct.
- (3) Both statement I and statement II are correct.
- (4) Both statement I and statement II are incorrect.

28. Which of the following is the **correct** floral formula of Poaceae family?

- (1) $Br \oplus \overset{\curvearrowright}{\underset{(3+3)}{P}} A_{3+3} \underline{G}_{(3)}$
- (2) $Br \% \overset{\curvearrowright}{\underset{2 \text{ or } 3 \text{ (lodicules)}}{P}} A_{3 \text{ or } 6} \underline{G}_1$
- (3) $\oplus \overset{\curvearrowright}{\underset{(5)}{K}} C_{(5)} A_5 \underline{G}_{(2)}$
- (4) $\% \overset{\curvearrowright}{\underset{(5)}{K}} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$

29. Which of the following is **not** the modified storage tap root?

- (1) Carrot
- (2) Sweet potato
- (3) Radish
- (4) Turnip

30. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Pedicel	(I)	Stalk of leaf
(B)	Bract	(II)	Swollen leaf base
(C)	Petiole	(III)	Stalk of flower
(D)	Pulvinus	(IV)	Reduced leaf

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-II, C-I, D-IV

31. The cell wall of kelps is made up of:

- (1) cellulose covered outside by algin.
- (2) cellulose and carrageen.
- (3) cellulose covered outside by lignin.
- (4) cellulose and suberin.

32. Spores in pteridophytes germinate to give rise to:

- (1) inconspicuous, multicellular, free-living prothallus.
- (2) conspicuous, multicellular, mostly non photosynthetic gametophytes.
- (3) large multicellular, photosynthetic sporophyte.
- (4) small unicellular prothallus.

33. Match **List-I** with **List-II**:

List-I		List-II	
(A)	<i>Chlamydomonas</i>	(I)	Moss
(B)	<i>Cycas</i>	(II)	Pteridophyte
(C)	<i>Selaginella</i>	(III)	Algae
(D)	<i>Polytrichum</i>	(IV)	Gymnosperm

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-II, C-I, D-IV

34. Read the following statements regarding bryophytes and select the **correct** option.

- A. Bryophytes lack true roots, stem and leaves.
- B. The main plant body is haploid.
- C. Sex-organs are unicellular.
- D. The sporophyte is not free-living.

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) B and C only
- (3) C and D only
- (4) A, B and D only

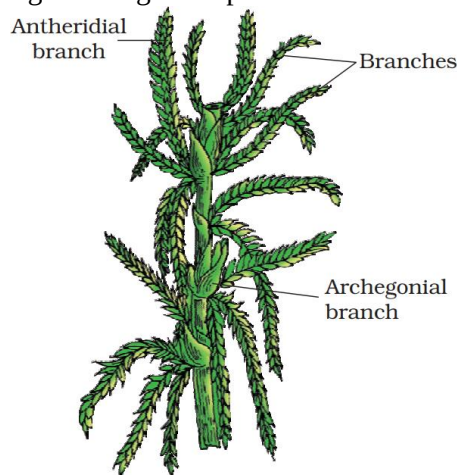
35. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The naked seeds are the characteristic feature of gymnosperms.

Reason R: In gymnosperms ovules are not enclosed by any ovary wall and remain exposed, both before and after fertilisation.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

36. The given diagram represents:



- (1) *Selaginella*
- (2) *Sphagnum*
- (3) Fern
- (4) *Salvinia*

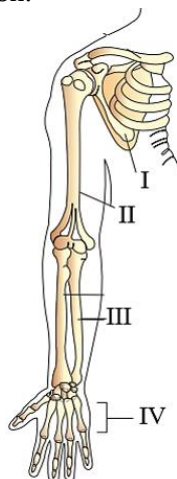
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Locomotion & Movement

SECTION-(IV) ZOOLOGY

37. Which of the following is a characteristic of cardiac muscles?

- (1) They are involuntary.
- (2) They are branched.
- (3) They are striated.
- (4) All of these

38. Refer to the given below diagram and choose the **correct** option.



- (1) I-Humerus
- (2) II-Scapula
- (3) III-Radius
- (4) IV-Metacarpals

39. Which of the following is an age related disorder caused by decreased levels of estrogen?

- (1) Gout
- (2) Osteoporosis
- (3) Rheumatoid arthritis
- (4) Myasthenia gravis

40. Read the following statements (A-E).

- A. In cartilaginous joints, the bones involved are joined together with the help of cartilages.
- B. Pelvic girdles meet ventrally to form the pubic symphysis containing fibrous cartilage.
- C. First seven pairs of ribs are called true ribs.
- D. A single U-shaped bone called hyoid is present at the base of the buccal cavity.
- E. The number of mitochondria are few in red muscle fibers.

How many of the above statements is/are **incorrect**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

41. Which of the following is **correct** w.r.t muscle fatigue?
- (1) It is due to the aerobic breakdown of glycogen.
 - (2) It is due to the breakdown of glycerol.
 - (3) It is due to glycogen formation.
 - (4) It is due to lactic acid accumulation.

42. Which of following bones is a part of the appendicular skeleton?
- (1) Clavicle
 - (2) Skull
 - (3) Sternum
 - (4) Ribs

43. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Microfilament	(I)	Skull
(B)	Visceral muscles	(II)	Amoeboid movement
(C)	Axial skeleton	(III)	Non-striated
(D)	Tetany	(IV)	Rapid spasm in muscle

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
 - (2) A-I, B-III, C-II, D-IV
 - (3) A-II, B-III, C-I, D-IV
 - (4) A-III, B-I, C-II, D-IV
44. During shortening of the muscle, the **X** get reduced, whereas the **Y** retain the length. Choose the option which fills the blank **correctly**.
- (1) X- I bands, Y- A bands
 - (2) X- A bands, Y- I bands
 - (3) X- A bands, Y- Z bands
 - (4) X- M bands, Y- I Bands
45. Given below are two statements:
- Statement I:** In the centre of each 'I' band is an elastic fibre called 'M' line which bisects it.
- Statement II:** The globular head of myosin has an active ATPase enzyme and has binding sites for ATP.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
46. Complete the analogy and choose the **correct** option w.r.t each human limb.
Wrist bones : 8 :: Palm bones : _____
- (1) 4
 - (2) 5
 - (3) 6
 - (4) 7
47. What would happen if ATP suddenly were **not** available after the sarcomere had started to shorten?
- (1) Cross bridges would not be able to detach from actin.
 - (2) Muscle would remain in a state of rigidity.
 - (3) Muscle would relax immediately.
 - (4) Both (1) and (2)
48. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Each organised skeletal muscle in our body is made of a number of fascicles.
Reason R: Fascicles are held together by a common collagenous connective tissue layer called myofibrils.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.

